

Compressibles Nozzle Analysis Project

This project analyzed the flow through both converging and converging-diverging nozzles using experimental data and computational fluid dynamics (CFD) simulations. Twelve tests were performed for each nozzle under different pressure ratios and flow conditions, with data collected from ten specific points along the nozzle wall during each test. MATLAB was utilized to process and organize the collected data, generating graphs to illustrate variations in pressure, Mach number, and mass flow rate. The analysis aimed to observe and compare different flow regimes, from no-flow to choked flow in the converging nozzle, and from no-flow to supersonic-isentropic flow in the converging-diverging nozzle. The study examined pressure and temperature changes across the nozzles, by comparing each result with its respective simulation output to assess its consistency with known nozzle theory. The findings included a detailed discussion and comparison of flow conditions and regimes for each test, as well as gave insight into the behavior of flow within varying nozzle types.

Nomenclature

A_i	=	reference nozzle area
A_e	=	exit nozzle area
<i>ANSYS</i>	=	computational fluid dynamics software used to model and run accurate simulations
<i>CN</i>	=	converging nozzle
<i>CDN</i>	=	converging-diverging nozzle
<i>CFD</i>	=	computational fluid dynamics
$\dot{m}$	=	mass flow rate
M	=	Mach number
<i>MATLAB</i>	=	matrix laboratory coding language software
<i>MFP</i>	=	mass flow parameter
P/P_0	=	pressure ratio
P_e	=	exit pressure
P_b	=	back pressure
P_i	=	static pressure
P_t	=	throat pressure
P_0	=	stagnation pressure
T_0	=	stagnation temperature
u	=	velocity of the flow
γ	=	specific heat of dry air

I. Introduction

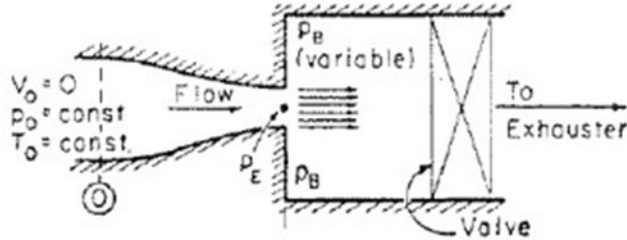
THIS project further teaches students to use, and become familiar with computational fluid dynamics (CFD) software. As high-performance computing becomes more common in the aerospace industry, being able to operate a program such as ANSYS has become increasingly valuable. Running CFD simulations allows engineers to create virtual models of real-world shapes and study the flow of fluids through them in a way that is both efficient and easy to repeat. These findings are important for the industry, as they typically produce products off which is most effective. For this analysis project, nozzle designs such as converging and converging-diverging nozzles were used to simulate the flow and examine the data.

A nozzle is a tube of varying cross-sectional areas (usually axisymmetric) aiming at increasing the speed of an outflow and controlling its direction and shape. Nozzle flow always generates forces associated to the change in flow momentum. In the simplest case of a rocket nozzle, relative motion is created by ejecting mass from a chamber backwards through the nozzle, with the reaction forces acting mainly on the opposite chamber wall, with a small contribution from nozzle walls. In jet propulsion, it is in the nozzle that thermal energy (or any other kind of high-pressure energy source) transforms into kinetic energy of the exhaust, and its associated linear momentum producing thrust. The isentropic model all along the nozzle is sufficient for a first-order analysis as the flow in a nozzle is very

rapid (and thus adiabatic to a first approximation) with very little frictional losses (because the flow is nearly one dimensional with a favorable pressure gradient, except if shock waves form and nozzles are relatively short). The nozzle is said to begin where the chamber diameter begins to decrease [1].

Figure 1: Schematic of a Converging Nozzle

Converging nozzles, as shown in Fig. 1, are used to accelerate the fluid in subsonic gas. A converging nozzle can only become supersonic at the exit stage; the speed increases monotonically along the nozzle. If a converging nozzle



is fed from a constant pressure and constant temperature chamber, the flow rate grows as the discharge pressure is being reduced, until the flow becomes sonic (choked) and the flow rate no longer changes with further decreasing in discharge-pressure (a set of expansion waves adjust the exit pressure to this lower discharge pressure). Except for old-time turbojets and military fighter aircraft, all commercial jet engines (after Concorde was retired) use converging nozzles discharging at subsonic speed (both the hot core stream and the colder fan stream). The back pressure (p_B)/temperature (T_B) is the driving factor that determines the flow condition in the nozzle. When $p_0 = p_B$, there is no flow through the nozzle. As p_B is reduced, the Mach number at the exit (p_E) keeps rising until the flow is choked ($ME = 1$). The condition at which choked flow occurs can be calculated using the isentropic relations, and for a $\gamma = 1.4$ when, $p_B/p_0 = 0.5283$ or, $T_B/T_0 = 0.8333$, the flow is said to be choked, i.e., the Mach number no longer increases and caps off at $M = 1$. The above conditions define the boundary between the non-choked and choked flow regimes, as shown in Fig. 2.

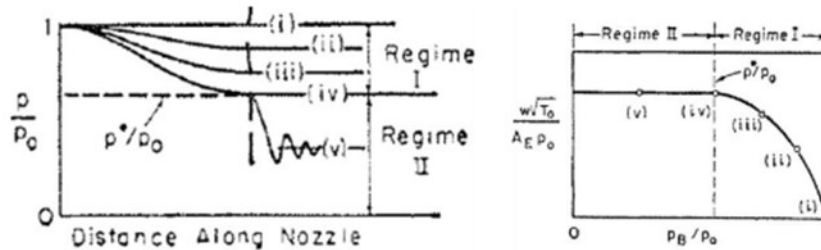


Figure 2: Flow Conditions and Regimes in a Converging Nozzle

Based on Fig. 2, the following are the flow conditions that can be observed in a converging nozzle:

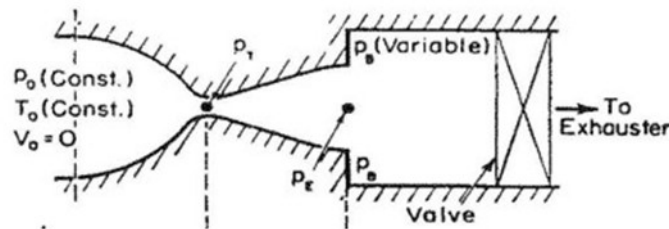
- i. No flow condition, where the back pressure is equal to the total pressure.
- ii. Subsonic flow, where the flow accelerates as area decreases and the pressure drops.
- iii. Subsonic flow, significantly higher acceleration and pressure drops.
- iv. Choked flow, where any pressure drop does not accelerate the flow.
- v. Choked flow, where the flow expands after the nozzle exit (considered non-isentropic).

Ramjets, scramjets, and rockets all use nozzles to accelerate hot exhaust to produce thrust as described by Newton's third law of motion. The amount of thrust produced by the engine depends on the mass flow rate through the engine, the exit velocity of the flow, and the pressure at the exit of the engine. The value of these three flow variables is all determined by the nozzle design. A nozzle is a relatively simple device, just a specially shaped tube through which hot gases flow. Ramjets and rockets typically use a fixed convergent section followed by a fixed divergent section for the design of the nozzle. This nozzle configuration is called a convergent-divergent, or CD, nozzle. In a CD nozzle, the hot exhaust leaves the combustion chamber and converges down to the minimum area, or throat, of the nozzle (Fig. 3). The throat size is chosen to choke the flow and set the mass flow rate through the system. The flow in the

throat is sonic which means the Mach number is equal to one in the throat. Downstream of the throat, the geometry diverges, and the flow is isentropically expanded to a supersonic Mach number that depends on the area ratio of the exit to the throat. The expansion of a supersonic flow causes the static pressure and temperature to decrease from the throat to the exit, so the amount of the expansion also determines the exit pressure and temperature. The exit temperature determines the exit speed of sound, which determines the exit velocity. The exit velocity, pressure, and mass flow through the nozzle determines the amount of thrust produced by the nozzle [2].

Figure 3: Schematic of a Convergent-Divergent Nozzle

As observed in the converging nozzle experiments, once the flow becomes choked, the velocity can no longer increase beyond the choked flow value. Based upon the equation, $du/u = -dA/A (1/1 - M^2)$, where u is the velocity,



A is the nozzle area, and M is the Mach number:

- a) At $M = 0$, static flow
- b) At $0 < M < 1$, as the area decreases, a proportional increase in the velocity of the flow will be observed
- c) At $M > 1$, any increase in area will produce a proportional increase in speed

Once the flow becomes choked, there are three possible flow conditions that can occur, subsonic isentropic flow (the flow decelerates after the choked condition), supersonic non-isentropic flow (where the flow accelerates supersonically, forms a normal shock, and decelerates subsonically after the shock), and supersonic isentropic flow (where the flow accelerates supersonically after the choked condition). Figure 4 shows the following seven profiles in the position versus pressure ratio plot:

1. Subsonic flow (never reaches choked condition).
2. Subsonic flow reaching choked condition, never reaching supersonic velocities (considered isentropic).
3. Subsonic flow reaching choked condition, the resulting supersonic flow results in a normal shock, then subsonic deceleration
4. Subsonic flow reaching choked condition, the resulting supersonic flow results in a normal shock, after the nozzle (considered isentropic in the nozzle).
5. Over-expanded flow.
6. The flow after the choked condition is supersonic through the nozzle, and no shock is formed.
7. Under-expanded flow.

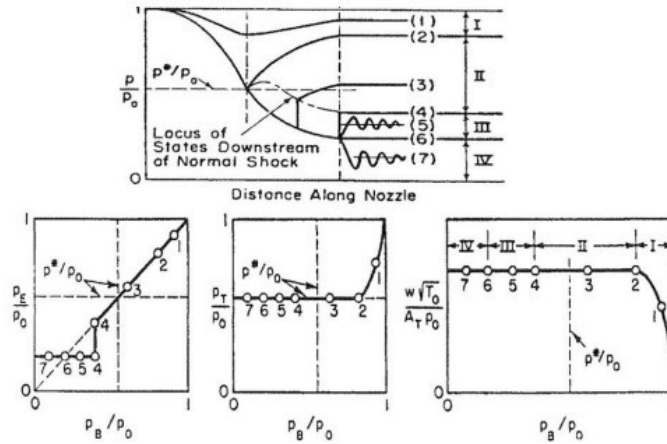


Figure 4: Flow Conditions and Regimes in a Convergent-Divergent Nozzle

Note that the isentropic relation given by the equation, $p^0/p_i = (1 + (\gamma - 1/2) M_i^2)^{\gamma/(\gamma-1)}$ can be used to determine the freestream Mach number. Here, P_0 is the inlet stagnation pressure, P_i is the static pressure at the wall for a given tap, i , M_i is the Mach number at the tap location, and γ is the specific heat of dry air. All pressures are absolute pressures.

II. Methodology

The procedure for this project included two main software, ANSYS and MATLAB. These applications required precision measurements to ensure accurate results, that way, the user could easily compare their values to known theories and formulas. The following methodology guides the reader through the process of the ANSYS CFD computations and the post-processing MATLAB script.

A. ANSYS Simulations

To begin, the user created models of each nozzle type in the ANSYS software's SpaceClaim tab. By creating these 2-D geometries, the user can adjust different parameters and their geometries at any point. Each sketch was oriented with a positive x-axis going to the right, the positive y-axis going up, and setting the units to meters. Using the dimensions given in the handout, the nozzle's shapes were created by connecting lines with the same geometry as the nozzle given in Appendix 2. In doing so, the user guarantees symmetry by making each side equal distance and applying length constraints. Then, the sketches were extruded into 3-D surfaces to complete their geometries. These surfaces play a key part in how the program knows where to simulate the airflow through.

The next step required the user to perform meshing. In this phase, the material was specified as a fluid, and the mesh size was set to be 0.001 meters to attain a high accuracy. Next, the user must create the mesh by selecting it in the toolbar, at which point, each side of the nozzle was labeled as its respective side, "Wall," "Inlet," and "Outlet." By doing so, the user ensured consistent boundary conditions when running simulations. The mesh was then saved for each nozzle geometry and can be found in Appendix 2.

For the setup section, the Fluent launcher was initialized using a double precision method, and the solver type was set to density-based. Next, inviscid flow and the energy equation were selected as models, and air was used for the ideal gas. According to the handout, the temperature was set to be 295 K, with different inlet pressures across each of the 12 runs. The outlet pressure was also adjusted for each new run, while the operating pressure stayed at zero. Using standard initialization to initialize the simulation, the inlet was chosen to be the starting point.

When the setup is fully initialized, a second order upwind is selected, along with the courant number being set to 5. Before running, the user was told to set the velocities and energy criteria to a tolerance of 1E-6. Each run was executed for 150 iterations for the converging nozzle and 300 iterations for the converging-diverging nozzle. After the simulations were complete, the results were collected and saved as files for later analysis in the MATLAB section. Multiple data types were collected during this stage such as static pressure and Mach number contours, which the user turned into excel files to be able to implement into the MATLAB code. The x-y plots for pressure and Mach distributions were exported for future processing within MATLAB. A contour of the static pressure and Mach number at each trial run are produced and can also be found in Appendix 2.

B. MATLAB Computations

During the MATLAB stage, the user took the data from the exported Excel files and organized it to more easily access it during the coding process. First off, the user created constants needed for computations throughout the code. Using the `readmatrix()` function, data files were imported from .xlsx (Excel) files. Using other similar commands in MATLAB, individual text files were merged into a single text file, which was then read into a matrix. Matrices for variables such as static pressure, back pressure, exit pressure, throat pressure, and mass flow rate were initialized, and using a loop function, processed the data from each trial run and added it to its respective table. This information was stacked and organized neatly so that it could be analyzed later.

Key calculations were performed to determine data like the pressure ratio, Mach number based on static and total pressures, and the mass flow parameter. Using these data points, four graphs were created for each different nozzle type, giving eight experimental plots. These graphs represented mass flow parameter versus back pressure ratio, Mach number versus axial nozzle distance, pressure ratio (P/P_0) versus axial nozzle distance, and mass flow parameter versus back pressure ratio (P_b/P_0).

After the experimental data was done processing, the user took a more detailed look at the data from the ANSYS computations. Using four excel files, the Mach and pressure data for both nozzle types were used to ease the tedious process of calculating the ANSYS outputs. Following the same process as before, the information was imported into MATLAB using the `readmatrix()` function, and then set into 3-D arrays. Meanwhile, another loop was created for variables such as Mach number and other pressure types to create more matrices. Next, twelve computational plots were generated using data from the new matrices. A more detailed explanation of each graph is presented in the Results and Discussion section, and the referenced MATLAB code is provided in Appendix 1.

III. Results and Discussion

The results and discussion section presents an analysis of experimental and computational results for both the converging and converging-diverging nozzle types. A multitude of graphs were generated to illustrate key relationships between different data sets. The experimental data, collected at specific points along the nozzle, is compared with continuous data points from ANSYS simulations to evaluate their accuracy and identify flow regimes. This segment examines how the data aligns with theoretical principles, highlights any inconsistencies between the experimental and computational results, as well as gives the reader insight into the flow behavior for both nozzle types.

A. Pressure Ratio (P/P_0) and Mach Number vs. Normalized Nozzle Distance for CN and CDN

Plots of pressure ratio and Mach number against the normalized distance along the nozzle were generated for all experimental runs and computational simulations. These plots included data from specific tap locations in the experimental runs, but continuously in the computational model. The experimental plots appeared less smooth due to using specific data points, while the computational plots were smoother and more detailed. This difference was mainly because the experimental data had only 10 tap points per run, while the computational results provided data at infinitely small intervals.

For the converging nozzle, these plots revealed clear transitions from the no-flow condition (Regime I), where the pressure ratio was at its highest, down to choked flow, where the Mach number reached a value of 1. The plots showed that as the back pressure decreased, the flow reached its choke condition and remained constant. It was not until the pressure was reduced beyond the critical value that indicated the flow was choked and was impossible to accelerate any further. Notably, the experimental data extended slightly beyond the nozzle exit, capturing real-world effects like downstream flow expansion. This resulted in higher Mach numbers and pressure ratios at the nozzle exit compared to the computational data, which was limited to within the nozzle and did not account for any dynamics past the nozzle's end (refer to Figures 5, 9). Pictured in Figure 2, this phenomenon matches what was expected of the flow.

For the converging-diverging nozzle, the plots showed subsonic flow accelerating through the converging section, until the flow hit Mach 1. Beyond the throat, the flow transitioned into Regime III, with supersonic expansion occurring in the diverging section. This behavior was captured in both experimental and computational data, though the experimental resolution was limited by 10 tap points. The computational data provided a continuous representation, capturing the supersonic flow in great detail. It also showed shockwave formations from Regime III – Regime IV, as the flow expanded, which also matched theoretical behavior (refer to Figures 6, 10).

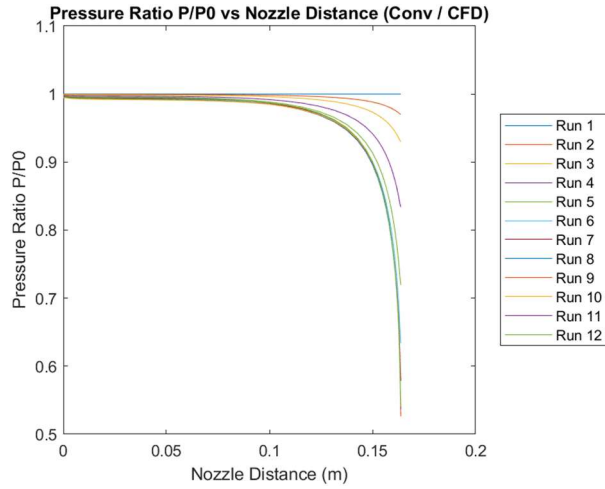


Figure 5: CFD P/P_0 vs Nozzle Distance (Converging)

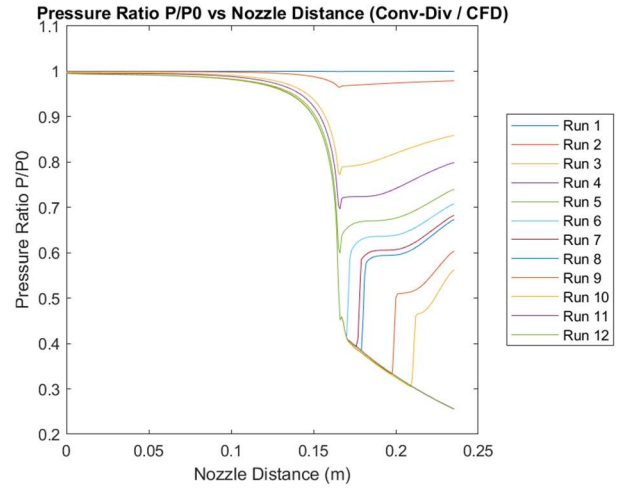


Figure 6: CFD P/P_0 vs Nozzle Distance (Converging-Diverging)

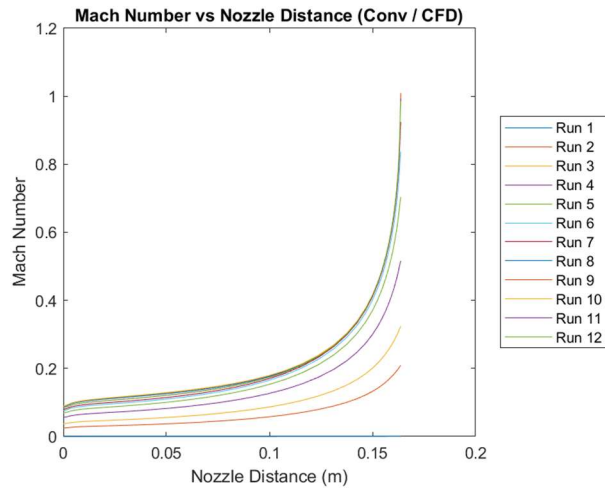


Figure 7: CFD Mach # vs Nozzle Distance (Converging)

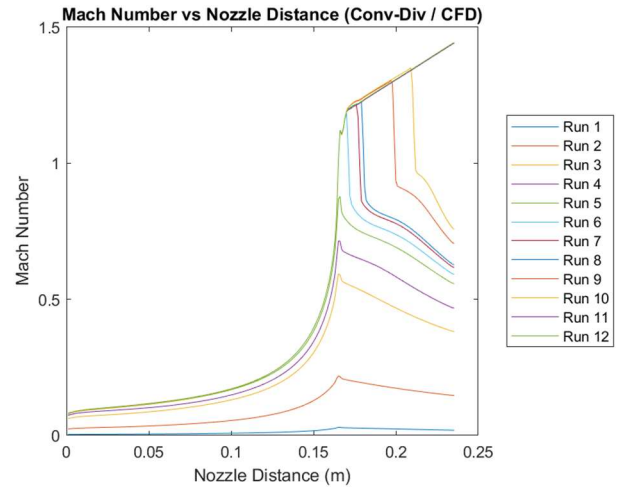


Figure 8: CFD Mach # vs Nozzle Distance (Converging-Diverging)

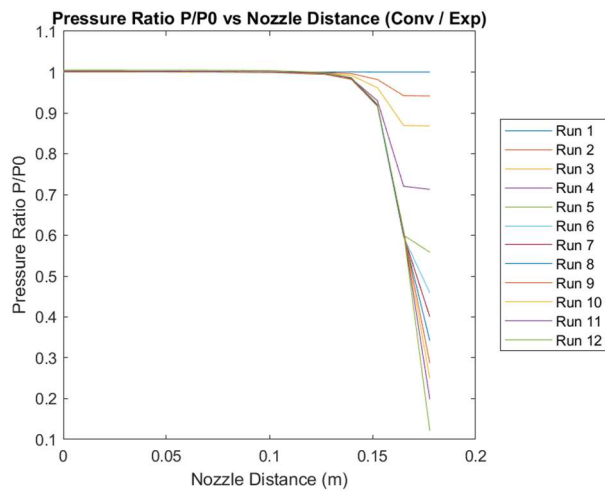


Figure 9: Exp P/P_0 vs Nozzle Distance (Converging)

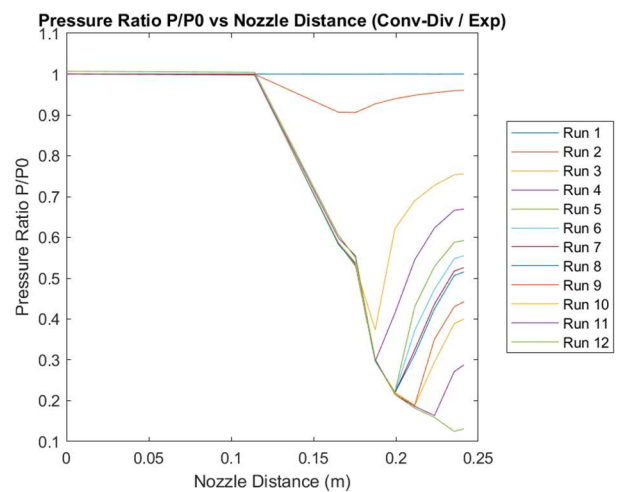


Figure 10: Exp P/P_0 vs Nozzle Distance (Converging-Diverging)

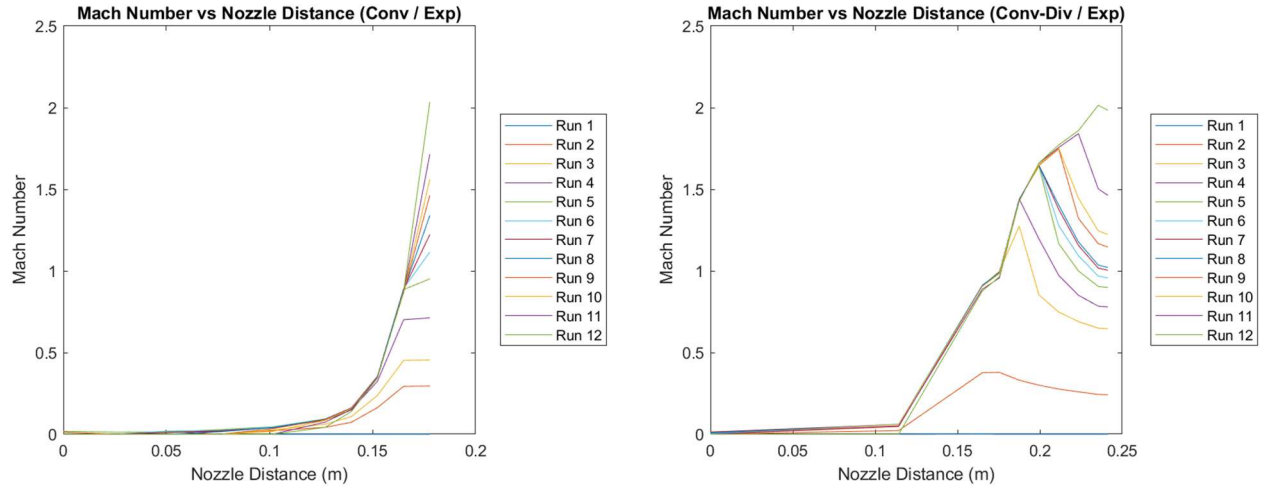


Figure 11: Exp Mach # vs Nozzle Distance (Converging) Figure 12: Exp Mach # vs Nozzle Distance (Converging-Diverging)

B. Comparison of Experimental and Computational Results for CN and CDN

Comparing the experimental to the computational results is crucial to assessing the accuracy of the simulation model. Experimental data, due to the discrete tap locations, showed a more skewed appearance at times, while the computational data, recorded continuously along the walls, was smoother and more detailed. Despite these differences, both data sets showed consistent trends across the range of runs.

In the case of the converging nozzle, the experimental data captured the transition to choked flow and then to subsonic flow, while computational data followed the flow to Mach 1 at the throat and its progression beyond the throat. The simulation provided an insight as to how the flow accelerated and decelerated in response to varying back pressures (refer to Figures 7 and 11). The experimental results extended slightly past the nozzle exit, capturing pressure taps that were not part of the computational model. In doing so, it illustrated real-world conditions that simulations may not be capable of replicating.

For the converging-diverging nozzle, experimental results showed subsonic flow in the converging section, a transition to choked flow at the throat, and then supersonic expansion in the diverging section. The computational results, with much better resolution, caught details of the supersonic expansion, including multiple shockwave formations and flow separations that were on par with theoretical expectations (refer to Figures 8 and 12). Overall, the experimental and computational data aligned well, confirming the simulations are consistent.

C. Mass Flow Parameter (MFP) vs. Back Pressure Ratio (P_b/P_0) for CN and CDN

The mass flow parameter was plotted against the back pressure ratio to study the effect of back pressure on the mass flow rate. The following equation was used to determine the mass flow parameter for both the converging and converging-diverging nozzle:

$$MFP = \frac{\dot{m}\sqrt{T_0}}{AP_0}$$

In the converging nozzle's plots, the trend showed a clear peak at choked flow (run 7), in accordance with the maximum mass flow rate. This peak indicated that once the flow reached the choked condition, the mass flow rate could not increase further, even when the back pressure started to decrease. In these plots, the no-flow condition was observed in Regime I, where the back pressure equaled the total pressure. Subsonic flow, also found in Regime I, was captured in runs 2-6, as the flow accelerated. In Figure 13, the experimental data showed that this characteristic peak was consistent with theoretical models, while the computational data provided a deeper view of a continuous representation.

For the converging-diverging nozzle, the MFP plot showed a similar trend, with the mass flow rate peaking at the choked condition. The plot demonstrated that the transition from subsonic to supersonic flow led to an increase in MFP, which remained stable in the supersonic flow regime. The experimental MFP plots for the converging-diverging nozzle differed from theoretical expectations in Figure 4, showing a hill-like shape with increasing and decreasing trends rather than leveling out. This behavior displays the more complex flow dynamics present in converging-diverging nozzles compared to simpler converging nozzles, seen in Figure 14.

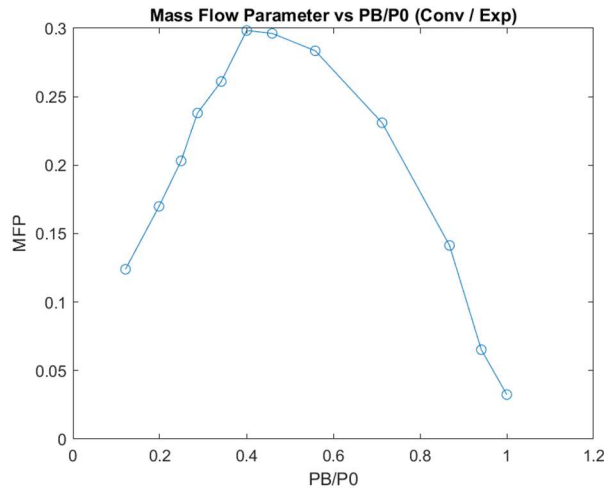


Figure 13: Exp MFP vs BPR (Converging)

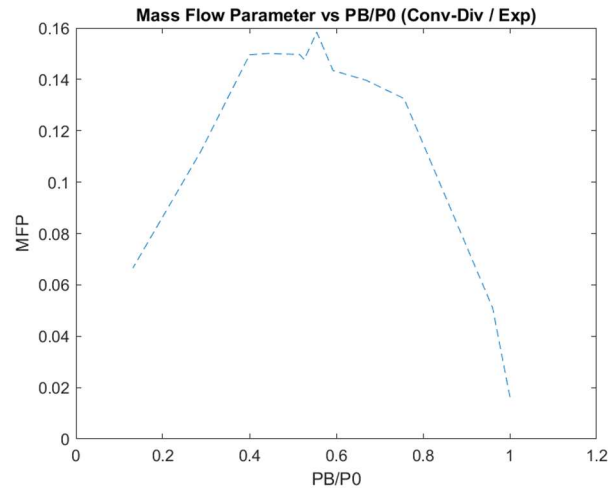


Figure 14: Exp MFP vs BPR (Converging-Diverging)

D. Throat and Exit Pressure Ratios (P_t/P_0 & P_e/P_0) vs. Back Pressure Ratio (P_b/P_0) for CDN

The plots of pressure ratios at the throat and exit against the back pressure ratio were generated for the converging-diverging nozzle. The results showed that as the back pressure increased, the throat pressure ratio approached the value for subsonic flow conditions and stayed relatively constant until choked flow occurred. This means the flow was subsonic, which matched Regime I. On the other hand, the exit pressure ratio followed a linear trend where it was lower at low back pressures and increased as back pressure increased, showing the transition from under-expanded to over-expanded flow conditions. Run 12 showed under-expanded flow, where the exit pressure was much lower than the surrounding pressure, fitting into flow Regime IV. These trends align with the theoretical model shown in Figure 2 and 4, and help identify flow regimes, such as fully and under-expanded flow in certain pressure conditions.

E. Flow Contours for Static Pressure and Mach Number for CN and CDN

Flow contours depicting static pressure and Mach number were created for all 12 computational runs. These visualizations allowed for an analysis of how flow conditions evolved along each nozzle. These are important to study to visualize what the flow regimes are for each varying pressure and Mach number.

For the converging nozzles, the contours displayed subsonic flow acceleration through the converging section, reaching Mach 1 at the throat, consistent with choked flow. The contour plots also illustrated pressure drop and velocity increase along the nozzle, following known fluid theorem. Across all runs, the contours revealed minimal variations in subsonic behavior, as the converging nozzle cannot achieve supersonic flow.

For the converging-diverging nozzles, the contours showed subsonic flow acceleration in the converging section, the formation of supersonic flow at the throat, and a decrease back to subsonic behavior in the diverging section. In runs with lower back pressure, the flow expanded supersonically in the diverging section. However, in some runs, the contours captured shockwaves in the diverging section, indicating over-expanded or under-expanded flows. These shockwaves were located closer to the exit in the over-expanded cases and further upstream in the under-expanded cases, aligning with theoretical ideals. All pressure and Mach contours referenced can be found in Appendix 2.

F. Tap Pressure Readings Analysis (P_b , P_e , P_t) for CN and CDN

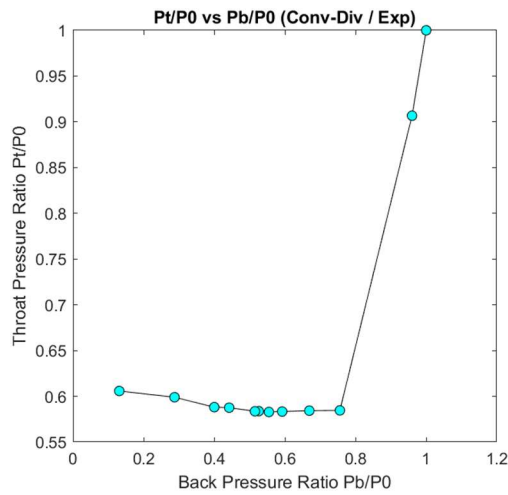
Pressure readings from specific taps were used in both experimental setups to capture key flow parameters:

- Tap 3: Used for measuring throat pressure (P_t) in the converging-diverging nozzle.
- Tap 9: Served as the exit pressure (P_e) condition for the experimental data.
- Tap 10: Used for measuring the back pressure (P_b) in both nozzles.

These readings were necessary for validating the computational results, ensuring that the experimental data matched theoretical models and expected flow regimes throughout the nozzle types. By accurately portraying these pressure conditions, the flow behavior in both the converging and converging-diverging nozzles was described effectively.

For the converging nozzle case, pressure taps 9 and 10 provided key measurements for analyzing the flow conditions. Tap 9, located at the nozzle's exit, measured the exit pressure. The experimental data showed slightly lower pressures at the exit compared to computational results, likely due to downstream flow interactions and increase in velocity. Tap 10, just downstream of the nozzle, recorded the back pressure, which defined the flow's boundary conditions. Experimental back pressure readings at tap 10 were often even lower than tap 9's readings, as predicted by the simulations, also shown in Figure 15.

For the converging-diverging pressure taps of 3, 9, and 10, the user found important differences between real and simulated flow behaviors. Tap 3, near the throat, measured the throat pressure. Experimental throat pressures were slightly lower than computational results, likely due to frictional forces or defects in the surfaces. Similarly, at taps 9 and 10, the recorded exit and back pressure seemed to be lower than the results from the simulations, which can be seen in Figure 16. This suggests that the flow expanded more after the nozzle exit than the computational model predicted it would. These happenings could result from further downstream expansions or weaker shockwaves. Experimental back pressures varied more due to environmental effects and external flow interactions, while computational data assumed stable conditions, as given in the handout's information. These distinctions showed that experiments included real-world effects, such as shocks and losses, which were simplified or excluded in simulation



runs.

Figure 15: Exp Throat Pressure vs Back Pressure

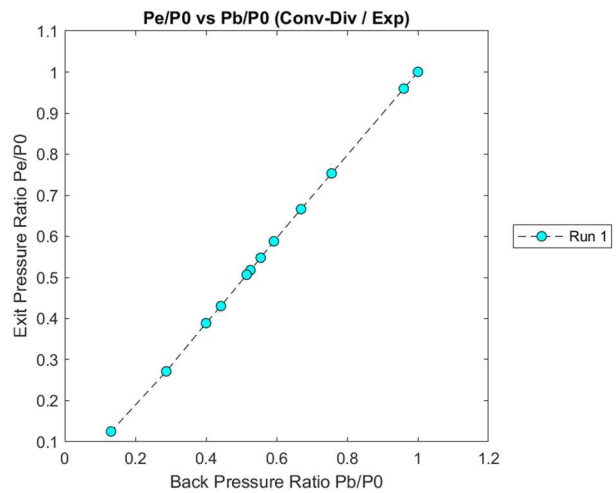


Figure 16: Exp Exit Pressure vs Back Pressure

IV. Conclusions

This project offered practical, hands-on experience with compressible aerodynamics, focusing on how airflow behaves in different nozzle types. By simulating subsonic and sonic flows using ANSYS software, students gained foundational skills essential for the aerospace engineering work field. The project also emphasized MATLAB programming, where data was processed and visualized through graphs, reinforcing the importance of problem-solving and complex coding required in engineering jobs. The study explored critical nozzle principles, such as the choked flow condition in converging nozzles, where the flow reached Mach 1 at the throat, but cannot exceed it. In contrast, converging-diverging nozzles showed the potential for supersonic flow in the diverging section under certain experimental conditions, due to an increase in velocity after the throat starts to expand. These findings displayed the importance of understanding various flow regimes, from subsonic flow to supersonic expansion, for designing systems involving nozzles. This project could also be a good study of how fluid flows act during certain pressure and temperatures, as well as when shockwaves are formed. The knowledge gained through this study is relevant to aerospace applications like jet engines, rockets, and wind tunnels, which every aerospace engineer will need to use in their future.

Appendix 1:

```
%Compressible Project 2 MATLAB Code
%Wesley Janes / wcj97

%Define constants given in handout
gamma = 1.4;
refnozzlearea = 3.17e-5;
stagtemp = 295;

%Experimental Data Read In's
ExpConv_1 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_1.txt');
ExpConv_2 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_2.txt');
ExpConv_3 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_3.txt');
ExpConv_4 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_4.txt');
ExpConv_5 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_5.txt');
ExpConv_6 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_6.txt');
ExpConv_7 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_7.txt');
ExpConv_8 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_8.txt');
ExpConv_9 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_9.txt');
ExpConv_10 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_10.txt');
ExpConv_11 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_11.txt');
ExpConv_12 = readmatrix('experimental_data_proj2\Converging_Nozzle\run_12.txt');
ExpConv_total =
cat(3,ExpConv_1,ExpConv_2,ExpConv_3,ExpConv_4,ExpConv_5,ExpConv_6,ExpConv_7,ExpConv_8,
ExpConv_9,ExpConv_10,ExpConv_11,ExpConv_12);

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readmatrix('experimental_data_proj2\ConvergingDiverging_Nozzle\run_10.txt');
ExpConvDiv_11 =
readmatrix('experimental_data_proj2\ConvergingDiverging_Nozzle\run_11.txt');
ExpConvDiv_12 =
readmatrix('experimental_data_proj2\ConvergingDiverging_Nozzle\run_12.txt');
ExpConvDiv_total =
cat(3,ExpConvDiv_1,ExpConvDiv_2,ExpConvDiv_3,ExpConvDiv_4,ExpConvDiv_5,ExpConvDiv_6,ExpConvDiv_7,ExpConvDiv_8,ExpConvDiv_9,ExpConvDiv_10,ExpConvDiv_11,ExpConvDiv_12);

unitchange = ExpConv_1(:,3)*0.0254;
unitchange2 = ExpConvDiv_1(:,3)*0.0254;
MFR_ExpConv = [];
MFR_ExpConvDiv = [];
NozzleA_ExpConv = [];
NozzleA_ExpConvDiv = [];
StagPress_ExpConv = [];
```

```

StagPress_ExpConvDiv = [];
StaticPress_ExpConv = [];
StaticPress_ExpConvDiv = [];
BackPress_ExpConv = [];
BackPress_ExpConvDiv = [];
ThroatPress_ExpConvDiv = [];
ExitPress_ExpConvDiv = [];

A = ones(10,12).*refnozzlearea;
for i = 1:1:12
    StaticPress_ExpConv = cat(2, StaticPress_ExpConv, ExpConv_total(:,5,i).*6894.76);
    StaticPress_ExpConvDiv = cat(2, StaticPress_ExpConvDiv,
    ExpConvDiv_total(:,5,i).*6894.76);

    BackPress_ExpConv = cat(2, BackPress_ExpConv, (ExpConv_total(10,5,i)).*6894.76);
    BackPress_ExpConvDiv = cat(2, BackPress_ExpConvDiv,
    (ExpConvDiv_total(10,5,i)).*6894.76);

    ExitPress_ExpConvDiv = cat(2, ExitPress_ExpConvDiv,
    ExpConvDiv_total(9,5,i).*6894.76);
    ThroatPress_ExpConvDiv = cat(2, ThroatPress_ExpConvDiv,
    ExpConvDiv_total(3,5,i).*6894.76);

    StagPress_ExpConv = cat(2, StagPress_ExpConv, ExpConv_total(:,6,i).*6894.76);
    StagPress_ExpConvDiv = cat(2, StagPress_ExpConvDiv,
    ExpConvDiv_total(:,6,i).*6894.76);

    MFR_ExpConv = cat(2, MFR_ExpConv, ExpConv_total(:,7,i).*14.5939);
    MFR_ExpConvDiv = cat(2, MFR_ExpConvDiv, ExpConvDiv_total(:,7,i).*14.5939);

    NozzleA_ExpConv = cat(2, NozzleA_ExpConv, ExpConv_total(:,4,i).*refnozzlearea);
    NozzleA_ExpConvDiv = cat(2, NozzleA_ExpConvDiv,
    ExpConvDiv_total(:,4,i).*refnozzlearea);
end

MFP_ExpConv = (MFR_ExpConv .* sqrt(stagtemp))./(pi*(0.0032^2).*StagPress_ExpConv);
MFP_ExpConvDiv = (MFR_ExpConvDiv .*
sqrt(stagtemp))./(pi*(0.0045^2).*StagPress_ExpConvDiv);
Mach_ExpConv = sqrt(5*(StagPress_ExpConv./StaticPress_ExpConv).^(1/3.5)-5);
Mach_ExpConvDiv = sqrt(5*(StagPress_ExpConvDiv./StaticPress_ExpConvDiv).^(1/3.5)-5);
BackPress0_ExpConv = BackPress_ExpConv./StagPress_ExpConv(1,:);
BackPress0_ExpConvDiv = BackPress_ExpConvDiv./StagPress_ExpConvDiv(1,:);

figure(1)
plot(BackPress0_ExpConv, MFP_ExpConv(1,:), '-o');
title('Mass Flow Parameter vs PB/P0 (Conv / Exp)')
xlabel('PB/P0')
ylabel('MFP')

figure(2)
plot(BackPress0_ExpConvDiv, MFP_ExpConvDiv(1,:), '--');
title('Mass Flow Parameter vs PB/P0 (Conv-Div / Exp)')
xlabel('PB/P0')
ylabel('MFP')

figure(3)
plot(unitchange, StaticPress_ExpConv ./ StagPress_ExpConv)
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Pressure Ratio P/P0 vs Nozzle Distance (Conv / Exp)')
xlabel('Nozzle Distance (m)')
ylabel('Pressure Ratio P/P0')

```

```

figure(4)
plot(unitchange2, StaticPress_ExpConvDiv ./ StagPress_ExpConvDiv)
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Pressure Ratio P/P0 vs Nozzle Distance (Conv-Div / Exp)')
xlabel('Nozzle Distance (m)')
ylabel('Pressure Ratio P/P0')

figure(5)
plot(unitchange, Mach_ExpConv)
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Mach Number vs Nozzle Distance (Conv / Exp)')
xlabel('Nozzle Distance (m)')
ylabel('Mach Number')

figure(6)
plot(unitchange2, Mach_ExpConvDiv)
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Mach Number vs Nozzle Distance (Conv-Div / Exp)')
xlabel('Nozzle Distance (m)')
ylabel('Mach Number')

figure(7)
plot(BackPress_ExpConvDiv ./ StagPress_ExpConvDiv(1,:), ExitPress_ExpConvDiv ./
StagPress_ExpConvDiv(1,:), '--ok', 'MarkerFaceColor', 'c')
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Pe/P0 vs Pb/P0 (Conv-Div / Exp)')
xlabel('Back Pressure Ratio Pb/P0')
ylabel('Exit Pressure Ratio Pe/P0')

figure(8)
plot(BackPress_ExpConvDiv ./ StagPress_ExpConvDiv(1,:), ThroatPress_ExpConvDiv ./
StagPress_ExpConvDiv(1,:), '-ok', 'MarkerFaceColor', 'c')
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Pt/P0 vs Pb/P0 (Conv-Div / Exp)')
xlabel('Back Pressure Ratio Pb/P0')
ylabel('Throat Pressure Ratio Pt/P0')

%CFD Data Read In's
Conv_Mach1 = readmatrix('Mach_Converging.xlsx','Sheet','C1','Range','A5:B44');
Conv_Mach2 = readmatrix('Mach_Converging.xlsx','Sheet','C2','Range','A5:B44');
Conv_Mach3 = readmatrix('Mach_Converging.xlsx','Sheet','C3','Range','A5:B44');
Conv_Mach4 = readmatrix('Mach_Converging.xlsx','Sheet','C4','Range','A5:B44');
Conv_Mach5 = readmatrix('Mach_Converging.xlsx','Sheet','C5','Range','A5:B44');
Conv_Mach6 = readmatrix('Mach_Converging.xlsx','Sheet','C6','Range','A5:B44');
Conv_Mach7 = readmatrix('Mach_Converging.xlsx','Sheet','C7','Range','A5:B44');
Conv_Mach8 = readmatrix('Mach_Converging.xlsx','Sheet','C8','Range','A5:B44');
Conv_Mach9 = readmatrix('Mach_Converging.xlsx','Sheet','C9','Range','A5:B44');
Conv_Mach10 = readmatrix('Mach_Converging.xlsx','Sheet','C10','Range','A5:B44');
Conv_Mach11 = readmatrix('Mach_Converging.xlsx','Sheet','C11','Range','A5:B44');
Conv_Mach12 = readmatrix('Mach_Converging.xlsx','Sheet','C12','Range','A5:B44');
Conv_Mach_total =
cat(3,Conv_Mach1,Conv_Mach2,Conv_Mach3,Conv_Mach4,Conv_Mach5,Conv_Mach6,Conv_Mach7,Con
v_Mach8,Conv_Mach9,Conv_Mach10,Conv_Mach11,Conv_Mach12);

ConvDiv_Mach1 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD1','Range','A5:B239');

```

```

ConvDiv_Mach2 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD2','Range','A5:B239');
ConvDiv_Mach3 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD3','Range','A5:B239');
ConvDiv_Mach4 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD4','Range','A5:B239');
ConvDiv_Mach5 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD5','Range','A5:B239');
ConvDiv_Mach6 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD6','Range','A5:B239');
ConvDiv_Mach7 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD7','Range','A5:B239');
ConvDiv_Mach8 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD8','Range','A5:B239');
ConvDiv_Mach9 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD9','Range','A5:B239');
ConvDiv_Mach10 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD10','Range','A5:B239');
ConvDiv_Mach11 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD11','Range','A5:B239');
ConvDiv_Mach12 =
readmatrix('Mach_Converging_Diverging.xlsx','Sheet','CD12','Range','A5:B239');
Conv_Div_Mach_total =
cat(3,ConvDiv_Mach1,ConvDiv_Mach2,ConvDiv_Mach3,ConvDiv_Mach4,ConvDiv_Mach5,ConvDiv_Mach6,ConvDiv_Mach7,ConvDiv_Mach8,ConvDiv_Mach9,ConvDiv_Mach10,ConvDiv_Mach11,ConvDiv_Mach12);

Conv_Press1 = readmatrix('Pressure_Converging.xlsx','Sheet','C1','Range','A5:B44');
Conv_Press1(:,2) = Conv_Press1(:,2)./(6894.76*29.21786);
Conv_Press2 = readmatrix('Pressure_Converging.xlsx','Sheet','C2','Range','A5:B44');
Conv_Press2(:,2) = Conv_Press2(:,2)./(6894.76*30.29996);
Conv_Press3 = readmatrix('Pressure_Converging.xlsx','Sheet','C3','Range','A5:B44');
Conv_Press3(:,2) = Conv_Press3(:,2)./(6894.76*31.59423);
Conv_Press4 = readmatrix('Pressure_Converging.xlsx','Sheet','C4','Range','A5:B44');
Conv_Press4(:,2) = Conv_Press4(:,2)./(6894.76*35.21712);
Conv_Press5 = readmatrix('Pressure_Converging.xlsx','Sheet','C5','Range','A5:B44');
Conv_Press5(:,2) = Conv_Press5(:,2)./(6894.76*40.78489);
Conv_Press6 = readmatrix('Pressure_Converging.xlsx','Sheet','C6','Range','A5:B44');
Conv_Press6(:,2) = Conv_Press6(:,2)./(6894.76*46.33998);
Conv_Press7 = readmatrix('Pressure_Converging.xlsx','Sheet','C7','Range','A5:B44');
Conv_Press7(:,2) = Conv_Press7(:,2)./(6894.76*50.87502);
Conv_Press8 = readmatrix('Pressure_Converging.xlsx','Sheet','C8','Range','A5:B44');
Conv_Press8(:,2) = Conv_Press8(:,2)./(6894.76*56.78238);
Conv_Press9 = readmatrix('Pressure_Converging.xlsx','Sheet','C9','Range','A5:B44');
Conv_Press9(:,2) = Conv_Press9(:,2)./(6894.76*64.35451);
Conv_Press10 = readmatrix('Pressure_Converging.xlsx','Sheet','C10','Range','A5:B44');
Conv_Press10(:,2) = Conv_Press10(:,2)./(6894.76*71.41687);
Conv_Press11 = readmatrix('Pressure_Converging.xlsx','Sheet','C11','Range','A5:B44');
Conv_Press11(:,2) = Conv_Press11(:,2)./(6894.76*84.91788);
Conv_Press12 = readmatrix('Pressure_Converging.xlsx','Sheet','C12','Range','A5:B44');
Conv_Press12(:,2) = Conv_Press12(:,2)./(6894.76*125.3289);
Conv_Press_total =
cat(3,Conv_Press1,Conv_Press2,Conv_Press3,Conv_Press4,Conv_Press5,Conv_Press6,Conv_Press7,Conv_Press8,Conv_Press9,Conv_Press10,Conv_Press11,Conv_Press12);

ConvDiv_Press1 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD1','Range','A5:B240');
ConvDiv_Press1(:,2) = ConvDiv_Press1(:,2)./(6894.76*29.56062);
ConvDiv_Press2 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD2','Range','A5:B240');
ConvDiv_Press2(:,2) = ConvDiv_Press2(:,2)./(6894.76*30.36138);
ConvDiv_Press3 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD3','Range','A5:B240');

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```

ConvDiv_Press3(:,2) = ConvDiv_Press3(:,2)./(6894.76*34.53745);
ConvDiv_Press4 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD4','Range','A5:B240');
ConvDiv_Press4(:,2) = ConvDiv_Press4(:,2)./(6894.76*37.08412);
ConvDiv_Press5 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD5','Range','A5:B240');
ConvDiv_Press5(:,2) = ConvDiv_Press5(:,2)./(6894.76*39.9443);
ConvDiv_Press6 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD6','Range','A5:B240');
ConvDiv_Press6(:,2) = ConvDiv_Press6(:,2)./(6894.76*41.62285);
ConvDiv_Press7 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD7','Range','A5:B240');
ConvDiv_Press7(:,2) = ConvDiv_Press7(:,2)./(6894.76*43.07477);
ConvDiv_Press8 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD8','Range','A5:B240');
ConvDiv_Press8(:,2) = ConvDiv_Press8(:,2)./(6894.76*43.65421);
ConvDiv_Press9 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD9','Range','A5:B240');
ConvDiv_Press9(:,2) = ConvDiv_Press9(:,2)./(6894.76*48.35942);
ConvDiv_Press10 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD10','Range','A5:B240');
ConvDiv_Press10(:,2) = ConvDiv_Press10(:,2)./(6894.76*51.77779);
ConvDiv_Press11 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD11','Range','A5:B240');
ConvDiv_Press11(:,2) = ConvDiv_Press11(:,2)./(6894.76*65.40394);
ConvDiv_Press12 =
readmatrix('Pressure_Converging_Diverging.xlsx','Sheet','CD12','Range','A5:B240');
ConvDiv_Press12(:,2) = ConvDiv_Press12(:,2)./(6894.76*119.0551);
ConvDiv_Press_total =
cat(3,ConvDiv_Press1,ConvDiv_Press2,ConvDiv_Press3,ConvDiv_Press4,ConvDiv_Press5,ConvDiv_Press6,ConvDiv_Press7,ConvDiv_Press8,ConvDiv_Press9,ConvDiv_Press10,ConvDiv_Press11,ConvDiv_Press12);

Converging_Mach = [];
Converging_Diverging_Mach = [];
Converging_Pressure = [];
Converging_Diverging_Pressure = [];

for i = 1:1:12
    Converging_Mach = cat(2,Converging_Mach, Conv_Mach_total(:,2,i));
    Converging_Diverging_Mach = cat(2,Converging_Diverging_Mach,
Conv_Div_Mach_total(:,2,i));

    Converging_Pressure = cat(2,Converging_Pressure,Conv_Press_total(:,2,i));
    Converging_Diverging_Pressure =
cat(2,Converging_Diverging_Pressure,ConvDiv_Press_total(:,2,i));
end

figure(9)
plot(Conv_Mach1(:,1), Converging_Mach)
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Mach Number vs Nozzle Distance (Conv / CFD)')
xlabel('Nozzle Distance (m)')
ylabel('Mach Number')

x_Converging_Diverging_Mach = ConvDiv_Mach1(:,1);
figure(10)
plot(x_Converging_Diverging_Mach, Converging_Diverging_Mach)
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Mach Number vs Nozzle Distance (Conv-Div / CFD)')
xlabel('Nozzle Distance (m)')

```

```

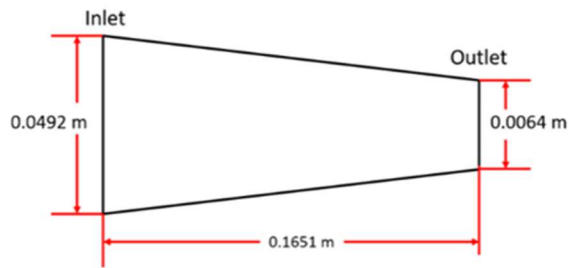
ylabel('Mach Number')

figure(11)
plot(Conv_Press1(:,1), Converging_Pressure)
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Pressure Ratio P/P0 vs Nozzle Distance (Conv / CFD)')
xlabel('Nozzle Distance (m)')
ylabel('Pressure Ratio P/P0')

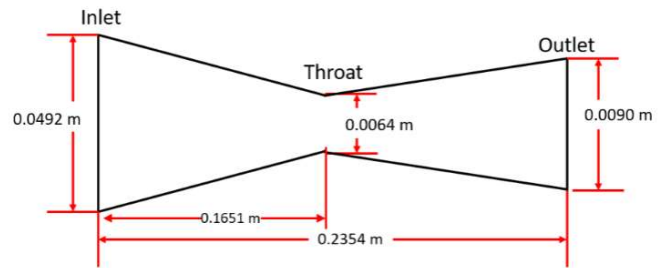
figure(12)
plot(ConvDiv_Press1(:,1), Converging_Diverging_Pressure)
legend('Run 1','Run 2','Run 3','Run 4','Run 5','Run 6','Run 7','Run 8', ...
'Run 9','Run 10','Run 11','Run 12','Location','eastoutside')
title('Pressure Ratio P/P0 vs Nozzle Distance (Conv-Div / CFD)')
xlabel('Nozzle Distance (m)')
ylabel('Pressure Ratio P/P0')

```

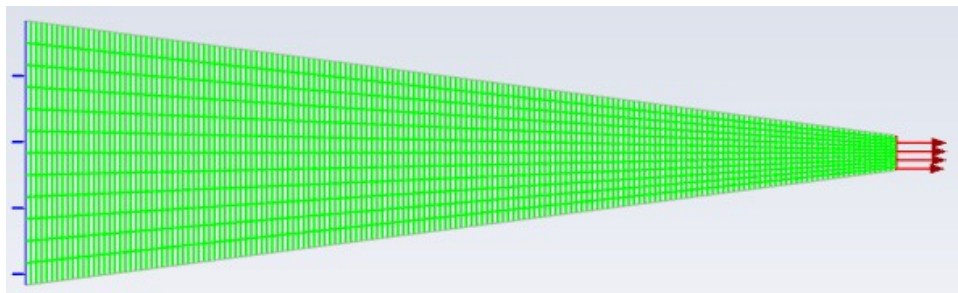

Appendix 2:



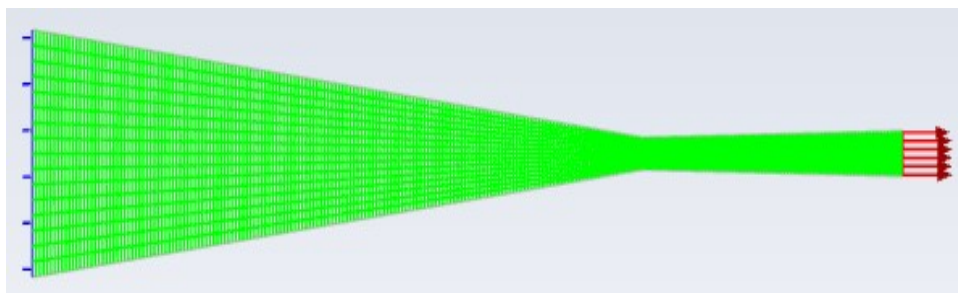
Converging Nozzle Geometry from Handout



Converging-Diverging Nozzle Geometry from Handout



Converging Nozzle Mesh from ANSYS

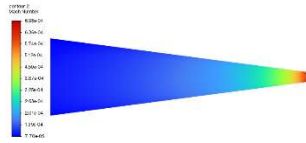


Converging-Diverging Nozzle Mesh from ANSYS

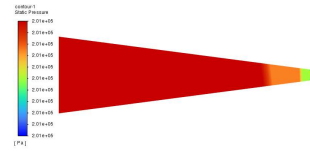
Converging Mach Contours (Left)

vs

Converging Pressure Contours (Right)



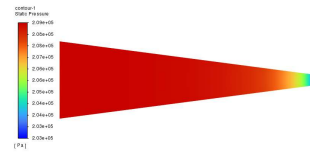
Run 1



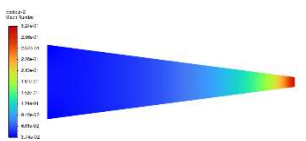
Run 1



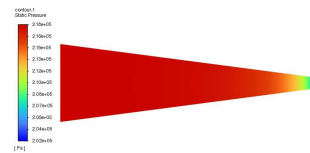
Run 2



Run 2



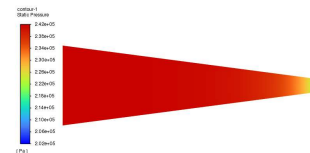
Run 3



Run 3



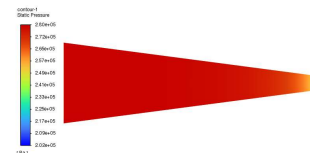
Run 4



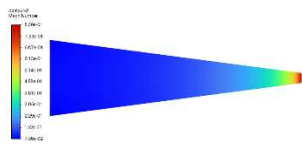
Run 4



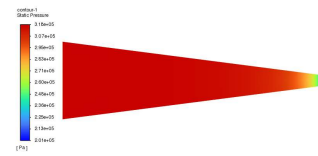
Run 5



Run 5



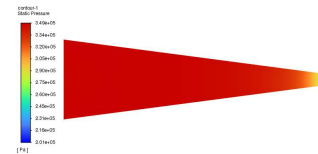
Run 6



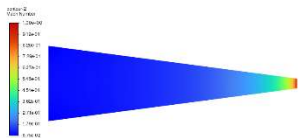
Run 6



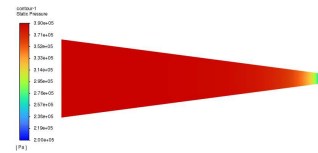
Run 7



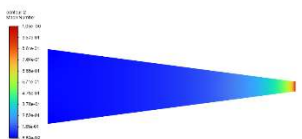
Run 7



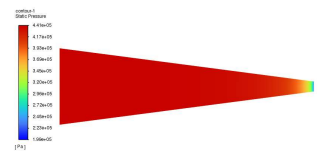
Run 8



Run 8



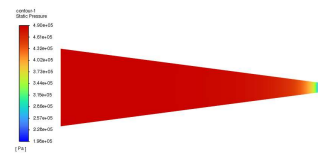
Run 9



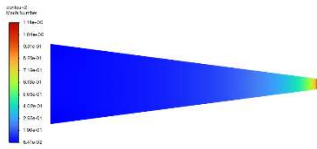
Run 9



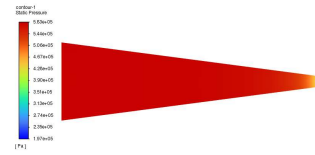
Run 10



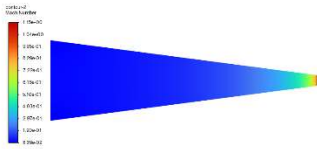
Run 10



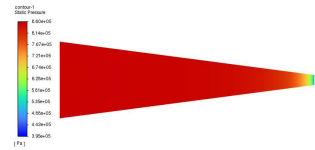
Run 11



Run 11



Run 12

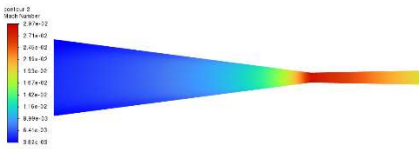


Run 12

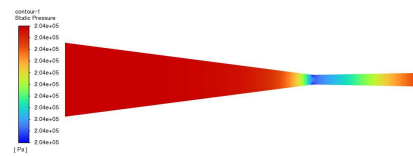
Converging-Diverging Mach Contours (Left)

vs

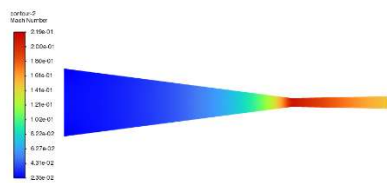
Converging-Diverging Pressure Contours (Right)



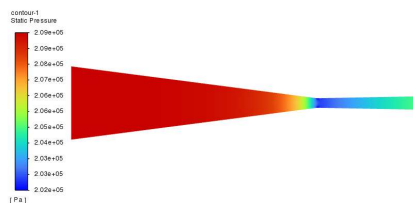
Run 1



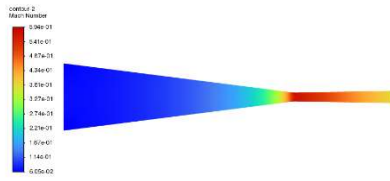
Run 1



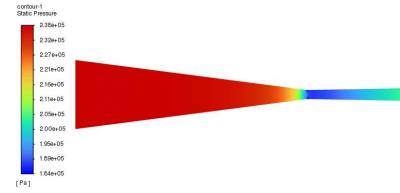
Run 2



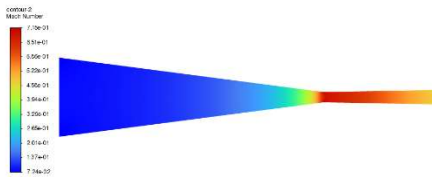
Run 2



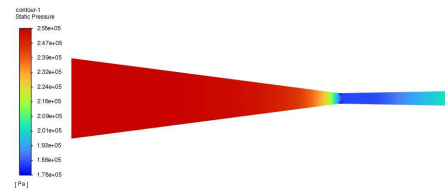
Run 3



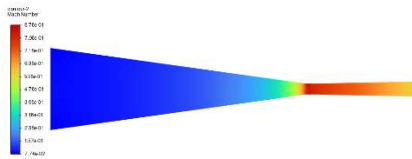
Run 3



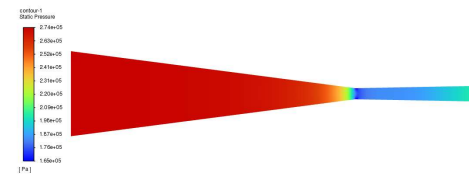
Run 4



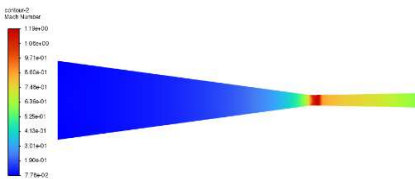
Run 4



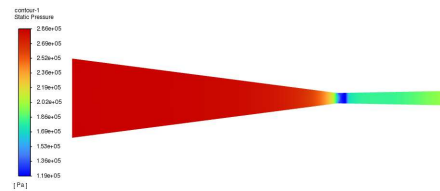
Run 5



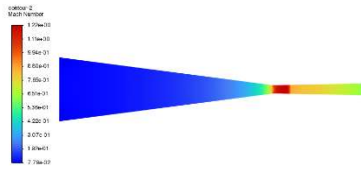
Run 5



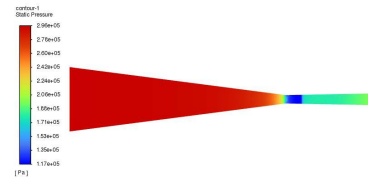
Run 6



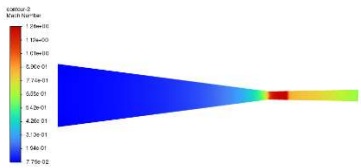
Run 6



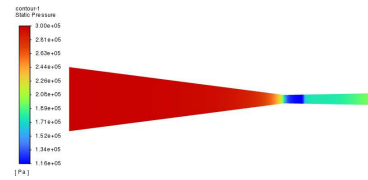
Run 7



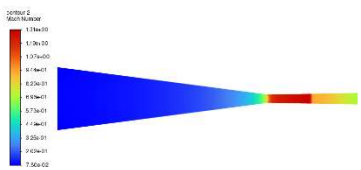
Run 7



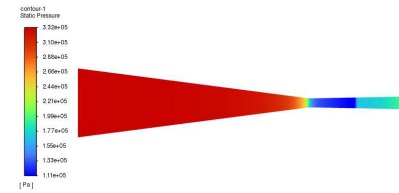
Run 8



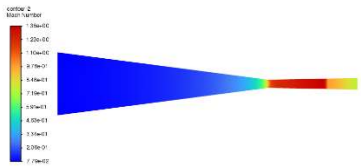
Run 8



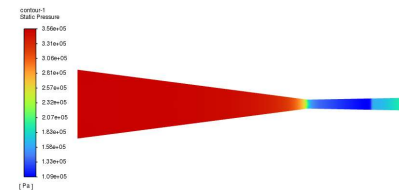
Run 9



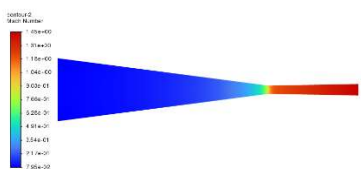
Run 9



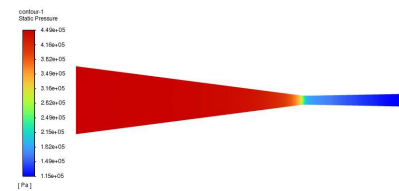
Run 10



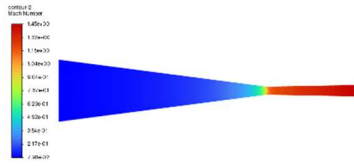
Run 10



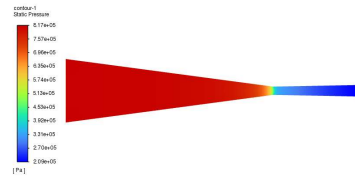
Run 11



Run 11



Run 12



Run 12

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- ² **MathWorks Inc.** MATLAB, Version R2022a. Natick, MA: The MathWorks Inc., 2022. Software.

Images

- ¹ Martinez, I., 'Nozzles', 1995. <http://webserver.dmt.upm.es/~isidoro/bk3/c17/Nozzles.pdf>
- ² NASA, 'Nozzle Design'. <https://www.grc.nasa.gov/www/k-12/airplane/nozzled.html>

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